



2009
TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 1

General Instructions

- Reading Time- 5 minutes
- Working Time – 2 hours
- Write using a black or blue pen
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown for every question.
- Begin each question on a fresh sheet of paper.

Total marks (84)

- Attempt Questions 1-7
- All questions are of equal value

Question 1 (12 marks) Begin a NEW page.**MARKS**

- a) Differentiate $\sin^{-1}(2x)$ **2**
- b) Find $\int x\sqrt{1+x^2} dx$ using the substitution $u = 1+x^2$ **2**
- c) Evaluate $\int_0^2 \frac{dx}{4+x^2}$ **3**
- d) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$ **1**
- e) Find the acute angle between the lines $3y = 2x + 8$ and $y = 5x - 9$ **2**
- f) Find the remainder when the polynomial $P(x) = x^3 - 4x$ is divided by $x + 3$. **1**
- g) Given that $\log_3 7 = m$, find an expression for $\log_3 21$. **1**

End of Question 1

Question 2 (12 marks) Begin a NEW page.**MARKS**

a) Solve $\cos \theta + \sqrt{3} \sin \theta = 1 \quad (0 \leq \theta \leq 2\pi)$

4

b) Solve $\frac{3x}{x-2} \leq 1$

3c) Find the coordinates of the point P which divides the interval AB externally in the ratio 1:2 where A is $(-2, 7)$ and B is $(1, 5)$.**2**

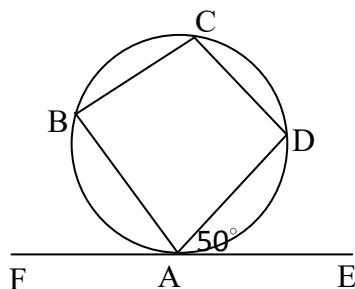
d) Solve $\cos 2\theta = \cos \theta \quad (0 \leq \theta \leq 2\pi)$

3**End of Question 2**

Question 3 (12 marks) Begin a NEW page.

MARKS

- a) ABCD is a cyclic quadrilateral and FAE is a tangent at A.
 $\angle DAE = 50^\circ$ and $BD \parallel FE$



- | | | |
|-----|--|---|
| i) | Calculate $\angle BAF$, giving reasons. | 2 |
| ii) | Calculate $\angle BCD$, giving reasons. | 2 |
| | | |
| b) | Solve the equation $2x^3 - 21x^2 + 42x - 16 = 0$ given that the roots form consecutive terms of a geometric sequence. | 3 |
| | | |
| c) | Evaluate $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sin x \cos^2 x \, dx$ using the substitution $u = \cos x$ | 3 |
| | | |
| d) | A packet of peas initially at 24°C is placed in a snap-freeze refrigerator which is maintained at an internal temperature of -40°C . Assuming Newton's Law of Cooling, the temperature T° of the peas will be given by $T = -40 + 64e^{kt}$, where k is a constant and t is the time in seconds. | |
| i) | If after 5 seconds the temperature of the packet is 19°C , Find the value of k . | 1 |
| ii) | How long will it take for the packet's temperature to reduce to 0°C ? Give your answer to the nearest second. | 1 |

End of Question 3

Question 4 (12 marks) Begin a NEW page.**MARKS**

- a) The chord PQ of the parabola $x^2 = 4y$ subtends a right angle at the origin O .

If the coordinates of P and Q are $(2p, p^2)$ and $(2q, q^2)$ respectively:

- i) Find the gradients of PO and QO . 1
- ii) Show $pq = -4$. 1
- iii) Find the equation of the locus of the midpoint M of PQ . 2

- b) A particle is moving along the x axis. Initially the particle is 2m to the right of the origin and moving with a velocity of 5 m/s .

- i) If the particle's acceleration is given by $\ddot{x} = 2x^3 + 2x$, show that $\dot{x} = x^2 + 1$. 3
- ii) Hence find an expression for x in terms of t . 2

- c) Water flows from a hole in the base of a cylindrical tank at a rate given by $\frac{dh}{dt} = -k\sqrt{h}$ where k is a constant and h is the depth of the water in mm at any time t minutes. 3

If the depth of the water falls from $100mm$ to $25mm$ in one minute, find how much longer it will take for the tank to empty.

End of Question 4

Question 5 (12 marks) Begin a NEW page.**MARKS**

- a) Prove by Mathematical Induction that $7^n + 5$ is divisible by 6, for positive integral values of n . **3**
- b) A missile is projected with a speed of 50m/s , at an angle of elevation of 45° . It is aimed at a tall building which is at a horizontal distance of 200m from the point of projection.
- i) By taking $g = 10\text{m/s}^2$ and neglecting air resistance, show that the equations of motion for this missile may be expressed as: **2**
- $$x = 25\sqrt{2}t \quad \text{and} \quad y = -5t^2 + 25\sqrt{2}t$$
- ii) Hence find the time of flight until the missile hits the building. **1**
- iii) Find the height of the missile when it hits the building. **1**
- c) Find m if: $\int_0^{\frac{\pi}{6}} \frac{\cos x \, dx}{1 + \sin x} = \log_e m$. **2**
- d) i) Find $\frac{d}{dx}[\tan^{-1} x + x]$ **1**
- ii) Hence evaluate $\int_0^1 \frac{x^2 + x + 2}{x^2 + 1} dx$ **2**
- (leave your answer in exact form)

End of Question 5

Question 6 (12 marks) Begin a NEW page.**MARKS**

a) For the curve with equation $y = \frac{x^2}{1-x^2}$

i) Show $\frac{dy}{dx} = \frac{2x}{(1-x^2)^2}$ **1**

ii) Find any stationary points and determine their nature. **2**

iii) Find the equations of any vertical asymptotes. **1**

iv) Find the equations of any horizontal asymptotes. **1**

v) Sketch the curve. **1**

b) A spherical balloon is being inflated and its volume increases at a constant rate of $50\text{mm}^3 / \text{s}$. **3**

At what rate is its surface area increasing when the radius is 20mm .

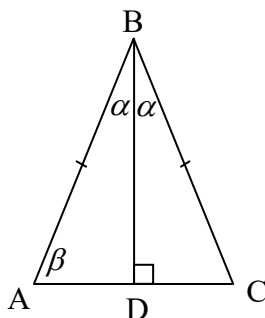
c) A particle moves such that $x = 3\cos 2t - 4\sin 2t$. **3**

Show this motion is Simple Harmonic Motion, and find the greatest speed of the particle.

End of Question 6

Question 7 (12 marks) Begin a NEW page.**MARKS**

a)



The triangle ABC is isosceles with $AB = BC$ and BD is perpendicular to AC . Let $\angle ABD = \angle CBD = \alpha$ and $\angle BAD = \beta$.

- i) Show $\sin \beta = \cos \alpha$ 1
- ii) Using the sine rule in $\triangle ABC$ show $\sin 2\alpha = 2 \sin \alpha \cos \alpha$ 2
- iii) Given that $0 \leq \alpha \leq \frac{\pi}{2}$, express the limiting sum of the geometric series $\sin 2\alpha + \sin 2\alpha \cos^2 \alpha + \sin 2\alpha \cos^4 \alpha + \dots$ in terms of α . 2
- .
- b) i) Evaluate: $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos^2 x \, dx$ Leave your answer in exact form. 2
- ii) Hence, if the curve $y = 2 + \cos x$ is rotated about the x axis from $\frac{\pi}{4} \leq x \leq \frac{\pi}{2}$ find the volume of the solid of revolution. 3
(leave your answer in exact form)
- c) Show without the use of a calculator: $\tan^{-1}(4) - \tan^{-1}\left(\frac{3}{5}\right) = \frac{\pi}{4}$. 2

End of Question 7**End of Examination**

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

$$\textcircled{1} a) \frac{d}{dx} \sin^{-1}(2x) = \frac{2}{\sqrt{1-4x^2}}$$

$$\begin{aligned} \text{b) } \int x \sqrt{1+x^2} \, dx & \quad u = 1+x^2 \\ \frac{du}{dx} &= 2x \\ & du = 2x \, dx \\ &= \frac{1}{2} \int \sqrt{u} \, du \\ &= \frac{1}{2} \times \frac{2}{3} u^{3/2} + C = \frac{1}{3} (1+x^2)^{3/2} + C \end{aligned}$$

$$\begin{aligned} c) \int_0^2 \frac{dx}{4+x^2} &= \frac{1}{2} \left[\tan^{-1} \left(\frac{x}{2} \right) \right]_0^2 \\ &= \frac{1}{2} \left[\tan^{-1} 1 - \tan^{-1} 0 \right] \\ &= \frac{1}{2} \left[\frac{\pi}{4} - 0 \right] = \frac{\pi}{8} \end{aligned}$$

$$d) \lim_{x \rightarrow 0} \frac{\sin 3x}{5x} = \lim_{x \rightarrow 0} \frac{3}{5} \left(\frac{\sin 3x}{3x} \right) = \frac{3}{5}$$

e) $y = \frac{2x}{3} + \frac{8}{3}$ $y = 5x - 9$
 $m_1 = \frac{2}{3}$ $m_2 = 5$

$$\tan \theta = \left| \frac{\frac{2}{3} - 5}{1 + \frac{2}{3} \times 5} \right| = 1$$
$$\theta = 45^\circ$$

f) $P(x) = x^3 - 4x$
 Remainder $= P(-3) = (-3)^3 - 4(-3)$
 $= -15$

$$\begin{aligned} \text{g) } \log_3 7 &= m & \log_3 21 \\ & & = \log_3 (7 \times 3) \\ & & = \log_3 7 + \log_3 3 \\ & & = m + 1 \end{aligned}$$

$$\begin{aligned} \textcircled{2} \text{ a) } \cos \theta + \sqrt{3} \sin \theta &= 1 \\ 2 \left[\frac{1}{2} \cos \theta + \frac{\sqrt{3}}{2} \sin \theta \right] &= 1 \\ \cos \left(\theta - \frac{\pi}{3} \right) &= \frac{1}{2} \\ \theta - \frac{\pi}{3} &= \frac{\pi}{3}, 2\pi - \frac{\pi}{3} \\ \theta &= 0, 2\pi, \frac{2\pi}{3} \end{aligned}$$

$$\begin{aligned} \text{b) } \quad & \frac{3x}{x-2} \leq 1 \\ & 3x(x-2) \leq (x-2)^2 \\ & 3x(x-2) - (x-2)^2 \leq 0 \\ & (x-2)(2x+2) \leq 0 \\ & -1 \leq x \leq 2 \\ \text{But } x & \neq 2 \quad \therefore -1 \leq x < 2 \end{aligned}$$

c) $A(-2, -7)$ $B(1, 5)$
 $-1:2$
 $x = \frac{-1 \times 1 + 2 \times -2}{-1 + 2}$ $y = \frac{-1 \times 5 + 2 \times 7}{-1 + 2}$
 $\therefore P(-5, 9)$

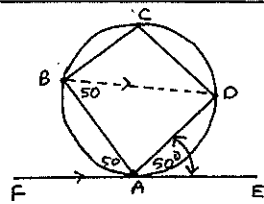
$$\begin{aligned} d) \quad \cos 2\theta &= \cos \theta \\ 2\cos^2 \theta - 1 &= \cos \theta = 0 \\ \text{let } m &= \cos \theta \\ 2m^2 - m - 1 &= 0 \\ (m-1)(2m+1) &= 0 \\ \cos \theta &= 1 & \cos \theta = -\frac{1}{2} \\ \theta &= 0, 2\pi, \frac{2\pi}{3}, \frac{4\pi}{3} \end{aligned}$$

③ a)

i) $\angle DBA = 50^\circ$ (alt. segment thm)
 $\therefore \angle BAF = 50^\circ$ (alt. L's, $BD \parallel FA$)

ii) $\angle BAD = 80^\circ$ ($\angle$ sum straight line)
 $\angle BCD = 100^\circ$ (opp L's cyclic quad)

b) $2x^3 - 21x^2 + 42x - 16 = 0$
Let roots be $\frac{\alpha}{\beta}, \alpha, \alpha\beta$
 $\therefore -\frac{16}{2} = \frac{\alpha}{\beta} + \alpha + \alpha\beta = \frac{21}{2}$
 $\frac{c}{a} = \frac{\alpha^2}{\beta} + \alpha^2 + \alpha^2\beta = 21$
 $-\frac{d}{a} = \alpha^3 = 8 \quad \therefore \alpha = 2$



③b) cont.

$$\frac{2}{\beta} + 2 + 2\beta = \frac{21}{2}$$
$$4\beta^2 - 17\beta + 4 = 0$$
$$(\beta - 4)(4\beta - 1) = 0$$
$$\beta = 4, \frac{1}{4}$$

$\therefore$ roots are $x = \frac{1}{2}, 2, 8$

$$\begin{aligned} \text{c) } \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sin x \cos^2 x \, dx & \quad u = \cos x \\ \frac{du}{dx} &= -\sin x \\ du &= -\sin x \, dx \\ x = \frac{\pi}{3} \quad u &= \frac{1}{2} \\ x = \frac{\pi}{2} \quad u &= 0 \\ \pi \left[-\frac{u^3}{3} \right]_0^{\frac{1}{2}} &= -\frac{1}{24} \end{aligned}$$

d) $T = -40 + 64e^{kt}$

i) $t=5$
 $T=19$

$$19 = -40 + 64e^{5k}$$
$$\frac{59}{64} = e^{5k}$$
$$k = \frac{1}{5} \ln\left(\frac{59}{64}\right) \div -0.016269$$

ii) $T=0$

$$0 = -40 + 64e^{kt}$$
$$\frac{40}{64} = e^{kt}$$
$$t = \frac{\ln\left(\frac{40}{64}\right)}{k} = 28.889$$

$\therefore$ takes approx 29 sec.

(14) a)

The diagram shows a coordinate system with x and y axes. A parabola is plotted with its vertex at the origin, labeled with the equation $x^2 = 4ay$. A chord is drawn between two points on the parabola, $P(2p, p^2)$ and $Q(2q, q^2)$. The midpoint of this chord is marked as M . A line segment connects the origin to M , and it is indicated that this segment is perpendicular to the chord PQ .

i) $m(PO) = \frac{p^2 - 0}{2p - 0} = \frac{p}{2}$ $m(OQ) = \frac{q^2 - 0}{2q - 0} = \frac{q}{2}$

ii) as $\angle POQ = 90^\circ$, then $m(PO) \times m(OQ) = -1$

$$\frac{p}{2} \times \frac{q}{2} = -1$$
$$pq = -4$$

iii) Midpoint $PQ = \left(\frac{2p+2q}{2}, \frac{p^2+q^2}{2} \right)$
 $= (p+q, \frac{p^2+q^2}{2})$
 $x = p+q$ $y = \frac{1}{2}(p^2+q^2)$
 $= \frac{1}{2}[(p+q)^2 - 2pq]$

$$y = \frac{1}{2} [x^2 - 2(-4)]$$
$$y = \frac{1}{2} (x^2 + 8)$$

b) $t=0$ $x=2$ $v=5$

i) $\ddot{x} = 2x^3 + 2x$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 2x^3 + 2x$$

$$\frac{1}{2} v^2 = \frac{2x^4}{4} + \frac{2x^2}{2} + C$$

$$\begin{aligned} v^2 &= x^4 + 2x^2 + c_1 \\ x=2 \quad v=5 \quad 25 &= 16 + 8 + c_1 \rightarrow c_1 = 1 \end{aligned}$$

$$\begin{aligned} v^2 &= x^4 + 2x^2 + 1 \\ &= (x^2 + 1)^2 \\ v &= \pm (x^2 + 1) \end{aligned}$$

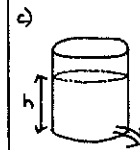
but as $x=2$
 $v=5$ $\therefore v = \dot{x} = x^2 + 1$

$$\text{ii) } v = \frac{dx}{dt} = x^2 + 1$$

$$\frac{dt}{dx} = \frac{1}{x^2 + 1}$$

$$\begin{aligned} t=0 & \quad x=2 \\ x &= \tan^{-1} x + C_2 \\ 0 &= \tan^{-1} 2 + C_2 \\ C_2 &= -\tan^{-1} 2 \end{aligned}$$

$$\begin{aligned}\therefore x &= \tan^{-1} x - \tan^{-1} 2 \\ \tan^{-1} x &= x + \tan^{-1} 2 \\ x &= \tan(x + \tan^{-1} 2) \\ &= \frac{\tan x + 2}{1 - 2 \tan x}\end{aligned}$$



$t=0$	$h=100$
$t=1$	$h=25$

$$\frac{dh}{dt} = -k\sqrt{h}$$

$$\frac{dh}{dt} = \frac{1}{-k\sqrt{h}} = \frac{h^{-1/2}}{-k}$$

$$0 = \frac{1}{-k} \cdot 2(100)^{1/2} + C$$

$$C = \frac{20}{K}$$

$$\therefore x = \frac{2\sqrt{h}}{-k} + \frac{20}{k}$$

$$I = \frac{2\sqrt{25}}{4K} + \frac{20}{K}$$

$k=10$

$$\therefore t = -\frac{\sqrt{h}}{5} + 2$$

4c) cont.

$$5t = -\sqrt{h} + 10$$

$$h = (10 - 5t)^2$$

empty when $h=0$ i.e. $0 = (10 - 5t)^2$
 $t = 2$

$\therefore$ takes a further 1 min to empty

5a) If $n=1$, $7^1 + 5 = 12$ which is divisible by 6

$\therefore$ True for $n=1$

Assume true for $n=k$

i.e. $7^k + 5 = 6M$ where M is an integer

$$7^k = 6M - 5$$

Aim to prove true for $n=k+1$

$$\text{i.e. } 7^{k+1} + 5 = 7 \cdot 7^k + 5$$

$$= 7(6M - 5) + 5$$

$$= 42M - 35 + 5$$

$$= 42M - 30$$

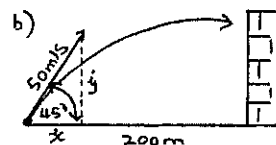
$$= 6(7M - 5)$$

$$= 6\theta \text{ where } \theta \text{ is an integer}$$

$\therefore$ If true for $n=k$, then true for $n=k+1$

Proved true for $n=1$, thus true for $n=1+1=2$

Hence by the principle of mathematical induction, true for all integer n , $n \geq 1$



$$t=0, x=0, y=0$$

$$\cos 45 = \frac{x}{50}$$

$$\dot{x} = 50 \cos 45 = \frac{50}{\sqrt{2}}$$

$$= 25\sqrt{2}$$

$$\ddot{x} = 0$$

$$\ddot{y} = -10$$

$$\dot{y} = -10t + C_2$$

$$t=0, \dot{x} = 25\sqrt{2}, \dot{y} = 25\sqrt{2}$$

$$\dot{x} = 25\sqrt{2}$$

$$\dot{y} = -10t + 25\sqrt{2}$$

$$x = 25t\sqrt{2} + C_3$$

$$y = -5t^2 + 25t\sqrt{2} + C_4$$

$$=0, x=0, y=0$$

$$x = 25t\sqrt{2}$$

$$y = -5t^2 + 25t\sqrt{2}$$

) Hits building when $x=200$

$$\text{i.e. } 200 = 25t\sqrt{2}$$

$$t = \frac{8}{\sqrt{2}} = 4\sqrt{2} \text{ sec}$$

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$$\text{iii) when } t = 4\sqrt{2} \quad y = -5(4\sqrt{2})^2 + 25\sqrt{2}(4\sqrt{2})$$

$$= 40$$

Hits 40m off ground

$$\begin{aligned} \text{c) } \int_0^{\pi/6} \frac{\cos x \, dx}{1 + \sin x} &= \left[\ln(1 + \sin x) \right]_0^{\pi/6} \\ &= \ln\left[1 + \sin\left(\frac{\pi}{6}\right)\right] - \ln[1 + \sin 0] \\ &= \ln 1.5 - 0 \\ &= \ln 1.5 \end{aligned}$$

$$\therefore m = 1.5$$

$$\begin{aligned} \text{d) i) } \frac{d}{dx} (\tan^{-1} x + x) &= \frac{1}{1+x^2} + 1 \\ &= \frac{2+x^2}{1+x^2} \end{aligned}$$

$$\begin{aligned} \text{ii) } \int_0^1 \frac{x^2 + x + 2}{x^2 + 1} \, dx &= \int_0^1 \frac{2+x^2}{1+x^2} + \frac{x}{1+x^2} \, dx \\ &= \left[\tan^{-1} x + x + \frac{1}{2} \ln(1+x^2) \right]_0^1 \\ &= \left[\tan^{-1} 1 + 1 + \frac{1}{2} \ln(2) \right] - \left[\tan^{-1} 0 + 0 + \frac{1}{2} \ln 1 \right] \\ &= \frac{\pi}{4} + 1 + \frac{1}{2} \ln 2 \end{aligned}$$

$$\text{6 a) } y = \frac{x^2}{1-x^2}$$

$$\begin{aligned} \text{i) } y' &= \frac{(1-x^2) \times 2x - x^2 \times (-2x)}{(1-x^2)^2} \\ &= \frac{2x - 2x^3 + 2x^3}{(1-x^2)^2} = \frac{2x}{(1-x^2)^2} \end{aligned}$$

ii) Stat pts when $y' = 0$

$$\text{i.e. } 2x = 0$$

$$x = 0$$

$$\text{at } x=0, y=0$$

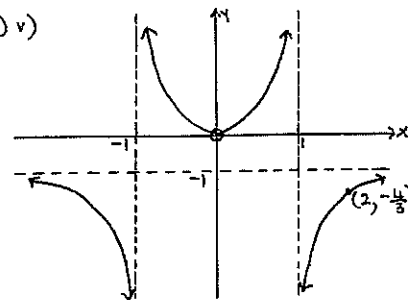
$\therefore$ min turning pt

$$D(0,0)$$

iii) $1-x^2=0 \Rightarrow$ vertical asymptotes at $x=1$ and $x=-1$

$$\begin{aligned} \text{iv) } \frac{-1}{1-x^2} &\Rightarrow y = \frac{x^2}{1-x^2} = -1 + \frac{1}{1-x^2} \\ \text{as } x \rightarrow 0, \frac{1}{1-x^2} &\rightarrow 0 \\ \therefore y &= \frac{x^2}{1-x^2} \rightarrow y = -1 \end{aligned}$$

6 a) v)



b)



$$\frac{dV}{dt} = 50 \text{ mm}^3/\text{s}$$

$$V = \frac{4}{3} \pi r^3$$

$$A = 4\pi r^2$$

$$\frac{dV}{dr} = 4\pi r^2$$

$$\frac{dA}{dr} = 8\pi r$$

$$\begin{aligned} \therefore \frac{dr}{dt} &= \frac{dr}{dV} \cdot \frac{dV}{dt} \\ &= \frac{1}{4\pi r^2} \times 50 \end{aligned}$$

$$\begin{aligned} \frac{dA}{dt} &= \frac{dA}{dr} \cdot \frac{dr}{dt} \\ &= 8\pi r \times \frac{25}{2\pi r^2} \end{aligned}$$

$$= \frac{25}{2\pi r^2}$$

$$\therefore \text{when } r=20 \quad \frac{dA}{dt} = 5 \text{ mm}^2/\text{s}$$

$$\text{c) } x = 3 \cos 2t - 4 \sin 2t$$

$$\dot{x} = -6 \sin 2t - 8 \cos 2t$$

$$\ddot{x} = -12 \cos 2t + 16 \sin 2t$$

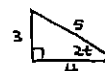
$$= -4(3 \cos 2t - 4 \sin 2t)$$

$$= -4x \text{ which is S.H.M.}$$

Greatest speed at centre of motion

$$\text{i.e. } 0 = 3 \cos 2t - 4 \sin 2t$$

$$\frac{\sin 2t}{\cos 2t} = \frac{3}{4} = \tan 2t$$



$$\therefore \text{Greatest speed} = \left| -6 \times \frac{3}{5} - 8 \times \frac{4}{5} \right| = 10$$

7 a)



i) In $\triangle ABD$,

$$\sin \beta = \frac{BD}{AB}$$

$$\cos \alpha = \frac{BD}{AB}$$

$$\therefore \sin \beta = \cos \alpha$$

ii) In $\triangle ABC$

$$\frac{AC}{\sin 2\alpha} = \frac{BC}{\sin \beta}$$

$$\sin 2\alpha = \frac{AC \sin \beta}{BC}$$

$$= \frac{2 DC \sin \beta}{BC} \quad (AC = 2DC)$$

$$= 2 \sin \alpha \sin \beta \quad (\sin \alpha = \frac{DC}{BC})$$

$$= 2 \sin \alpha \cos \alpha \quad (\text{from (i)})$$

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$$\text{iii) } 0 \leq x \leq \frac{\pi}{2} \quad \sin 2x + \sin 2x \cos 2x + \sin 2x \cos^2 2x + \dots$$

$$\rightarrow \text{GP } a = \sin 2x \quad r = \cos^2 2x$$

$$\begin{aligned} \therefore S_{\infty} &= \frac{a}{1-r} = \frac{\sin 2x}{1-\cos^2 2x} = \frac{2 \sin 2x \cos 2x}{\sin^2 2x} \\ &= \frac{2 \cos 2x}{\sin 2x} \\ &= 2 \cot 2x \end{aligned}$$

$$\begin{aligned} \text{b) i) } \int_{\pi/4}^{\pi/2} \cos^2 x \, dx &= \frac{1}{2} \int_{\pi/4}^{\pi/2} (\cos 2x + 1) \, dx \\ &= \frac{1}{2} \left[\frac{\sin 2x}{2} + x \right]_{\pi/4}^{\pi/2} \\ &= \frac{1}{2} \left[\left(\frac{1}{2} \sin \pi + \frac{\pi}{2} \right) - \left(\frac{1}{2} \sin \frac{\pi}{2} + \frac{\pi}{4} \right) \right] \\ &= \frac{1}{2} \left[\frac{\pi}{2} - \frac{1}{2} - \frac{\pi}{4} \right] = \frac{\pi}{8} - \frac{1}{4} \end{aligned}$$

$$\text{ii) } y = 2 + \cos x$$

$$y^2 = 4 + 4 \cos x + \cos^2 x$$

$$V = \pi \int_{\pi/4}^{\pi/2} 4 + 4 \cos x + \cos^2 x \, dx$$

$$= \pi \left[4x + 4 \sin x + \frac{1}{3} \cos^3 x \right]_{\pi/4}^{\pi/2} + \pi \int_{\pi/4}^{\pi/2} \cos^2 x \, dx$$

$$= \pi \left[\left(2\pi + 4 \sin \frac{\pi}{2} \right) - \left(\pi + 4 \sin \frac{\pi}{4} \right) \right] + \pi \left(\frac{\pi}{8} - \frac{1}{4} \right)$$

$$= \pi \left[\frac{9\pi}{8} + \frac{15}{4} - 2\sqrt{2} \right] \text{ units}^3$$

$$\text{c) LHS} = \tan^{-1}(4) - \tan^{-1}\left(\frac{3}{5}\right)$$

$$\text{Let } \tan^{-1} 4 = x \quad \tan^{-1}\left(\frac{3}{5}\right) = y$$

$$\tan x = 4$$

$$\tan y = \frac{3}{5}$$

$$\tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

$$= \frac{4 - \frac{3}{5}}{1 + 4 \times \frac{3}{5}}$$

$$= 1$$

$$x - y = \tan^{-1}(1)$$

$$\therefore \tan^{-1} 4 - \tan^{-1}\left(\frac{3}{5}\right) = \frac{\pi}{4}$$

THE END!